

Autonomous Traffic-Lightless Intersection Control

1. Safety requirements

Collision Prevention

- Two conflicting vehicles shall never occupy the intersection simultaneously.

Mutual Exclusion

- Opposite unsafe directions shall never receive permission together.

Emergency Handling

- The system shall enter emergency mode if:
 - communication fails
 - sensor failure occurs
 - vehicle behavior becomes inconsistent

Deadlock Prevention

- The system shall never reach a state where all vehicles wait forever.

Safe Distance

- Vehicles shall maintain minimum separation distance.

Fail Safe Operation

- In case of controller failure:
 - All vehicles must stop before entering the intersection.

Priority Safety

- Emergency vehicles shall receive priority safely.

Safety_Requirements

<<requirement>> SR1	
+ID: SR1	
+Title: Collision Avoidance	
+Input: Vehicle trajectories, Conflict area occupancy	
+Output: Safe crossing permissions, Collision prevention	
+Description: No two vehicles shall occupy the same conflict area or collision block simultaneously.	

<<requirement>> SR2	
+ID: SR2	
+Title: Cross-Collision Point Safety	
+Input: Vehicle arrival timing, Crossing paths	
+Output: Conflict-free scheduling	
+Description: Vehicles from conflicting directions shall not reach the same collision point at the same time.	

<<requirement>> SR3	
+ID: SR3	
+Title: Permission Before Entry	
+Input: Vehicle crossing request	
+Output: Entry permission or denial	
+Description: Vehicles shall only enter the intersection after receiving crossing permission from the intersection coordination unit.	

<<requirement>> SR4	
+ID: SR4	
+Title: Minimum Vehicle Separation	
+Input: Vehicle positions, Vehicle speeds	
+Output: Safe spacing enforcement	
+Description: Vehicles traveling in the same lane shall maintain safe separation distance.	

<<requirement>> SR5	
+ID: SR5	
+Title: Safe Turning Speed	
+Input: Turning direction, Vehicle speed	
+Output: Reduced turning speed command	
+Description: Vehicles performing left or right turns shall reduce speed to safe turning limits.	

<<requirement>> SR6	
+ID: SR6	
+Title: Communication Failure Safety	
+Input: Communication status	
+Output: Stop command	
+Description: A disconnected vehicle without valid permission shall stop before the stop line.	

<<requirement>> SR7	
+ID: SR7	
+Title: Sensor Failure Safe Mode	
+Input: Sensor reliability status	
+Output: Safe mode activation	
+Description: If roadside or vehicle sensors become unreliable, the system shall switch to a safe operational mode.	

2. Timing requirements

Response Timing

- The controller shall respond to a vehicle request within a bounded time.

Crossing Constraints

- A vehicle shall clear the intersection within a maximum crossing time.

Communication Timing

- Vehicle status updates shall be transmitted periodically.

Timeout Handling

- If a vehicle does not respond within a timeout interval, the controller shall revoke permission.

Scheduling

- Only one conflicting traffic flow may be active at a time.
- Safe switching delay shall exist between permissions.

Timing_Requirements

<<requirement>> TR1
+ID: TR1 +Title: Controller Response Time +Input: Vehicle position, Vehicle speed, Vehicle request message +Output: Crossing permission, Speed guidance +Description: The intersection coordination unit shall compute and send crossing permission within 2 seconds after receiving vehicle data.

<<requirement>> TR2
+ID: TR2 +Title: Maximum Vehicle Waiting Time +Input: Vehicle queue information, Traffic density +Output: Scheduled crossing order +Description: No vehicle shall wait indefinitely for crossing permission during normal operation.

<<requirement>> TR3
+ID: TR3 +Title: Safe Intersection Clearance Time +Input: Reserved conflict area, Vehicle trajectory +Output: Safe crossing completion +Description: Vehicles shall clear reserved conflict areas within the assigned crossing time window.

<<requirement>> TR4
+ID: TR4 +Title: Communication Timeout Detection +Input: V2I/V2X communication status +Output: Timeout warning, Stop command +Description: The system shall detect V2I or V2X communication failure within 3 seconds.

<<requirement>> TR5
+ID: TR5 +Title: Emergency Reaction Time +Input: Obstacle detection, Emergency vehicle signal +Output: Permission cancellation, Emergency priority assignment +Description: The controller shall cancel unsafe permissions immediately after detecting an obstacle, accident, or emergency vehicle.

<<requirement>> TR6
+ID: TR6 +Title: Sensor Update Timing +Input: Camera data, Radar data, Vehicle sensor data +Output: Updated environment model +Description: Vehicle sensors, roadside cameras, and radars shall provide periodic environment updates in real time.

<<requirement>> TR7
+ID: TR7 +Title: Safe Switching Delay +Input: Current traffic flow state +Output: Delayed conflicting permission +Description: The controller shall maintain a safe temporal separation between conflicting vehicle movements.

